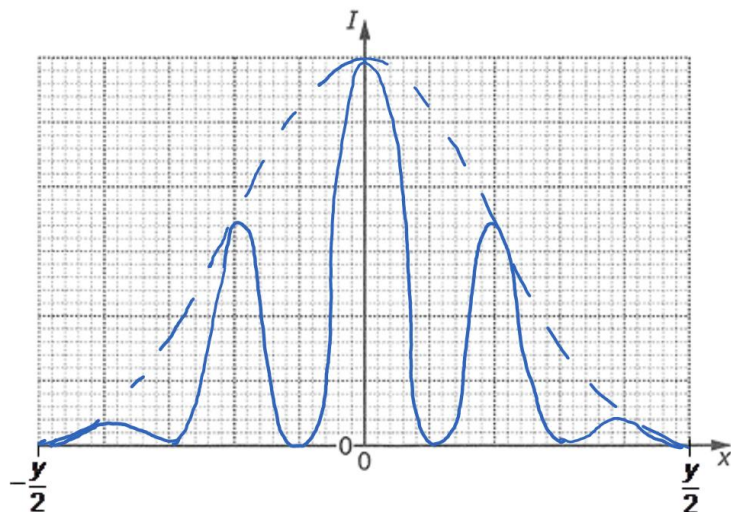


4ai	Diagram must show: (1) Wavefront curving into geometric shadow with some spreading (2) Wavelength unchanged (minus 1 mark if less than 4 wavelengths)
4aii	Angle of diffraction increases or More spreading of the wave.
4bi	$\sin \theta = \frac{\lambda}{b}$ $\sin \theta = (4.5 \times 10^{-7}) / (0.50 \times 10^{-3})$ Let width be c. $\tan \theta = 0.5 \text{ c } / 2.5$ $\text{c} = 4.500 \text{ mm} = 4.5 \text{ mm}$ Accept small angle approximations.
4ii1	<u>Coherent waves (from laser) diffracted from the slits will overlap and interfere.</u> When waves meet (in phase) with phase difference of $2n\pi$ rad OR with path difference of $n\lambda$, where $n = 0, \pm 1, \pm 2, \dots$, <u>constructive interference</u> occurs and (maximum intensity is observed). And at the points where the waves (meet anti-phase) with phase difference of $(2n - 1)\pi$ rad OR with path difference of $(n + \frac{1}{2}) \lambda$, <u>destructive interference</u> occurs and (minimum intensity) is observed.
4ii2	fringe separation, $x = \lambda D / a$ $= (4.5 \times 10^{-7} \times 2.5) / 1.25 \times 10^{-3}$ $= 9.00 \times 10^{-4} = 0.90 \text{ mm}$ Number of bright fringes = $4.5 / 0.90$ $= 5$

4bii3



Intensity of maxima varies according to single diffraction envelope.

Correct number of fringes

Reasonable smooth curve with equal fringe separation

5a

Increasing current causes increasing magnetic flux (linkage) through the coil.
(By Faraday's Law), increasing magnetic flux (linkage) induces e.m.f. in the coil.